

Chapter 63: Neurological Disorders — Revision Notes

Williams Obstetrics 26e, pp. 2933–2977 of the PDF. High-yield numbers are in **bold**. The four tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 5 figures are included from the PDF.

Big picture: Most neurological disorders predate pregnancy and most women do well, but new neurological symptoms must be separated from pregnancy complications (especially **preeclampsia**). Neurovascular disorders caused **>7% of US maternal deaths** (2007–2016). Psychiatric disorders can present with cognitive and neuromuscular findings.

1. Central Nervous System Imaging

- **CT and MR imaging are safe in pregnancy** (Ch 49).
 - **CT:** rapid diagnosis; excellent for **recent hemorrhage**.
 - **MR imaging is often preferred:** demyelinating disease, **AVMs**, congenital/developmental anomalies, **posterior fossa** lesions, spinal cord disease.
- Advanced pregnancy: **lateral tilt with a wedge under one hip** (prevents hypotension; ↓ aortic pulsation artifact).
- **Cerebral angiography** (usually femoral) is a valuable adjunct for cerebrovascular disease. Fluoroscopy gives more radiation but can be shielded.
- PET and fMRI have not been evaluated in detail in gravidas.

2. Headache

- **17%** of US adults aged 18–44 reported a severe headache/migraine in the past 3 months (**24%** of nonpregnant women that age).
- In pregnant women given a neurology consult for headache:
 - **Two thirds primary** → **migraine in >90%** of these.
 - **One third secondary** → **more than half from hypertensive disorders**.
- **20% of postpartum ED visits** were for headache (Parkland/UTSW).
- Primary headaches are more common than secondary in pregnancy. **Migraine** is the type most affected by pregnancy hormones.

Table 63-1. Headache Classification

Primary	Secondary
Migraine	Trauma
Tension-type	Vascular disorders
Trigeminal cephalgia	Substance abuse
Other	Infection
	Disorders of homeostasis
	Craniofacial disorders
	Psychiatric disorders

International Headache Society (2018).

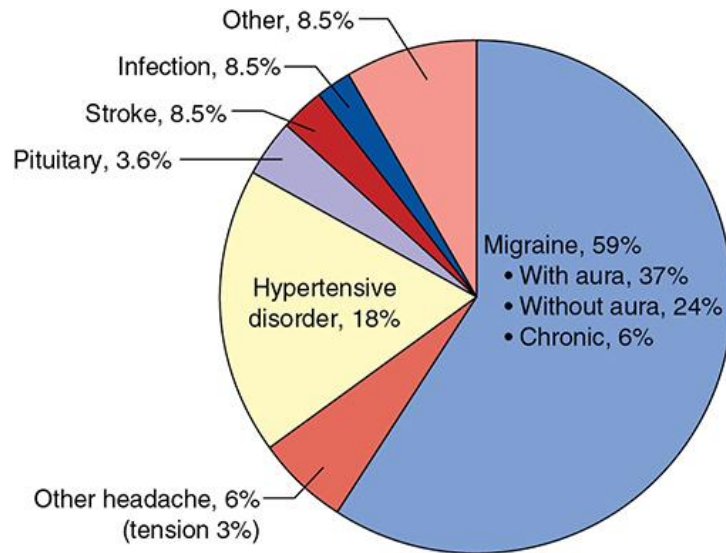


Figure 63-1. Causes of headache in 140 pregnant women seen by neurology: migraine 59%, hypertensive disorder 18%, stroke/infection/other 8.5% each.

Tension headache

- **Most frequent cause of all headaches**; episodic or chronic.
- Muscle tightness, mild–moderate pain in the **back of the neck and head**, lasting hours.
- **No neurological disturbance or nausea** → distinguishes it from migraine.
- Treatment: rest, massage, heat/ice, NSAIDs, mild tranquilizers. **Admission seldom needed.** Chronic → **amitriptyline**.

Migraine

- Periodic, sometimes incapacitating: severe headache + **autonomic dysfunction**.
- **IHS three types**:
 1. **Without aura** (formerly "common"): unilateral throbbing headache, nausea/vomiting or photophobia.
 2. **With aura** (formerly "classic"): same, preceded by visual scotoma or hallucinations. **One third** of patients. May be averted by taking medication at the first premonitory sign.
 3. **Chronic**: migraine on **≥15 days/month for >3 months**.
- Annual prevalence **15% women vs 6% men**; another 5% of women have "probable migraine" (all criteria but one).
- Begins in childhood, peaks in adolescence, wanes with age. Linked to hormone levels.
- Pathophysiology: neuronal dysfunction → ↓ cortical blood flow, activation of vascular/meningeal nociceptors and **trigeminal sensory neurons**. **Not primarily a vascular disorder**.
- **Migraine with aura in young women** → ↑ **ischemic stroke**. Also linked to asthma, anxiety/depression, other pain disorders.

Migraine in pregnancy

- **Most migraineurs improve** in pregnancy. New-onset migraine (usually **with aura**) can appear in pregnancy. New neurological symptoms → **full evaluation**.
- Incidence **185 per 100,000 deliveries**.
- Associated risks in gravid migraineurs:

Outcome	Risk
Stroke	16-fold
Myocardial infarction	5-fold
Heart disease, VTE, preeclampsia/gestational hypertension	2-fold each

- Also ↑ preeclampsia, gestational hypertension, preterm birth and **low birthweight**.

Management

- Mild headaches respond to oral therapy in **50–70%**.
- First line: **acetaminophen or an NSAID + antiemetic**.
- Severe → **parenteral therapy**: IV hydration, parenteral antiemetics, **triptans**. **Opioids only for the most recalcitrant**. Multitargeted therapy is usually needed.
- **Magnesium sulfate 2 g** has gained favor, but a metaanalysis showed **no benefit**.
- ⚠ **Ergotamine derivatives are avoided** (potent vasoconstrictors, **uterotonic**).
- **Triptans** (5-HT_{1B/1D} agonists): relieve headache, nausea and vomiting; oral, injection, rectal or nasal. Best **combined with NSAIDs**.
 - **Sumatriptan** has the most experience; **not a major teratogen**. One study found neurodevelopmental differences (emotionality, activity) at 36 months.
 - ⚠ **Avoid chronic NSAIDs, especially late** → **ductus arteriosus constriction** and ↓ fetal urine output.
- Peripheral nerve blocks help some women.
- **Prophylaxis** (frequent migraines):

Drug	Daily dose
Amitriptyline	10–175 mg
Propranolol	40–120 mg
Metoprolol	25–100 mg

- **CGRP-receptor monoclonal antibodies** (approved since 2018): **not evaluated in pregnancy**. Botulinum toxin A ↓ chronic migraine by ~2 days/month.

Table 63-2. Medications for Treatment of Acute Migraine in Pregnancy

Class	Agents
Simple analgesics	Acetaminophen; NSAIDs: naproxen, ibuprofen
5-HT receptor agonists	Triptans: oral, nasal, parenteral
Dopamine-receptor antagonists	Metoclopramide, prochlorperazine, chlorpromazine
Antiemetics	Ondansetron, metoclopramide
Other	Opioids; transcranial magnetic stimulation

5-HT = 5-hydroxytryptamine (serotonin); NSAIDs = nonsteroidal antiinflammatory drugs.

The text calls triptans "5-HT_{1B/2D}" agonists; the correct receptor subtypes are 5-HT_{1B/1D}.

Cluster headache

- Rare. Severe **unilateral lancinating pain to face and orbit**, lasting **15–180 min**, with autonomic symptoms and agitation.
- **Pregnancy does not change severity**. Avoid tobacco and alcohol.
- Acute: **100% oxygen + sumatriptan 6 mg SC**. Recurrent → **verapamil** prophylaxis.

Pearl: A "new" or "worst" headache in pregnancy, or one with focal signs, is secondary until proven otherwise; check the blood pressure first.

3. Seizure Disorders

- **Second most common** neurological condition in pregnancy (after headache).
- 3 million US adults had active epilepsy (2015); fewer than half were seizure-free in the past year.
- **Epilepsy caused 5% of UK maternal deaths** (2011–2013).

Pathophysiology

- Seizure = transient signs/symptoms from abnormal excessive or synchronous neuronal activity.
- Causes in young adults: head trauma, **alcohol/drug withdrawal**, infection, tumors, biochemical abnormalities, **AVMs**. Search for these with a new-onset seizure in pregnancy.
- **Idiopathic epilepsy is a diagnosis of exclusion.**

Classification (ILAE 2017)

Onset	Features
Focal	One localized area. Trauma, abscess, tumor, perinatal factors (lesion rarely shown). Awareness intact or impaired.
— without dyscognitive features	Start in one body region and spread ipsilaterally; tonic → clonic; sensory, autonomic or psychological change. Cognition intact, rapid recovery.
— with dyscognitive features	Often preceded by aura ; then behavioral arrest or motionless stare; automatisms (picking, lip smacking).
Generalized	Both hemispheres ; ± aura, then abrupt loss of consciousness; motor or nonmotor. Strong hereditary component.
— tonic-clonic	LOC → tonic rigidity → clonic jerking. Gradual recovery; confusion for hours.
— absence (petit mal)	Brief LOC without muscle activity; immediate recovery.
Unknown	Unwitnessed onset; subgrouped motor/nonmotor/unclassified.

Preconceptional counseling

- **Folic acid 0.4 mg/d**, starting **≥1 month before conception** → **increase to 4 mg/d** when a woman taking antiepileptic drugs becomes pregnant.
- Goal: **monotherapy with the least teratogenic drug** at the **lowest effective dose**.
- Consider **withdrawal if seizure-free ≥2 years**.

Pregnancy complications

- Main risks: **fetal malformations and convulsions**.
- **Seizures ↑ in only 20–30%** of gravidas today. **Seizure-free ≥9 months before conception** → likely to stay seizure-free.
- Causes of subtherapeutic levels:
 - Nausea/vomiting, slower GI motility, **antacids** ↓ absorption
 - Hypervolemia (offset by ↓ protein binding)
 - **Hepatic enzyme induction**, placental drug metabolism
 - ↑ **GFR** → faster clearance
 - **Stopping medication** out of teratogenicity fear
- ↓ Seizure threshold: sleep deprivation, **hyperventilation and pain in labor**.
- **Status epilepticus = medical emergency** (more likely with refractory epilepsy). In benzodiazepine-refractory status, IV **levetiracetam, fosphenytoin or valproate** worked equally; only **half** stopped seizing by 60 min.
- Small ↑ in: miscarriage, hemorrhage, hypertensive disorders, abruption, preterm birth, FGR, cesarean.

- **Maternal death rate 10-fold higher.** ↑ Postpartum depression.
- **Children of epileptic mothers: 10% risk of seizure disorder.**

Malformations

- **Untreated epilepsy is not linked to ↑ malformations;** the risk comes from certain **anticonvulsants**.
- **Monotherapy < polytherapy;** risk rises with **each drug added**.
- **Phenytoin, phenobarbital:** major malformations **2–3×** baseline.
- **Valproate:** most potent, **dose-dependent, 4–8×** risk; also **lower cognitive performance**.
- **Lamotrigine and levetiracetam carry the lowest malformation risk** (metaanalysis of 31 studies).

Table 63-3. Teratogenic Effects of Common Anticonvulsant Medications

Drug (Brand Name)	Abnormalities Described	Affected	Embryofetal Risks ^a
Valproate (Depakote)	Neural-tube defects , clefts, cardiac anomalies; associated developmental delay; ADHD	7–10% with monotherapy; higher with polytherapy	<i>(blank in print)</i>
Phenytoin (Dilantin)	Fetal hydantoin syndrome: craniofacial anomalies, fingernail hypoplasia, growth deficiency, developmental delay, cardiac anomalies, clefts	5–11%	Yes
Carbamazepine; oxcarbazepine (Tegretol; Trileptal)	Fetal hydantoin syndrome, as above, spina bifida	2–5%	Yes
Phenobarbital	Clefts, cardiac anomalies, urinary tract malformations	6–20%	Suggested
Lamotrigine (Lamictal)	Increased risk for clefts (registry data)	Up to 2% (4- to 10-fold higher than expected)	Suggested
Topiramate (Topamax)	Clefts	2–3% (15- to 20-fold higher than expected)	Suggested
Levetiracetam (Keppra)	Theoretical: skeletal abnormalities; impaired growth in animals	1–3%	Suggested

ADHD = attention deficit hyperactivity disorder. ^aRisk categories from Briggs, 2022, and others. The printed table leaves the valproate risk cell empty; the text calls valproate a "particularly potent teratogen".

Memory hook: "Valproate = Very bad (NTDs, IQ ↓); Lamotrigine/Levetiracetam = Least risk."

Management in pregnancy

- Goal = **seizure prevention:** treat nausea/vomiting, avoid triggers, **stress compliance**, fewest drugs at lowest dose.
- **Serum drug levels are unreliable** (altered protein binding); free levels aren't widely available; **monitoring does not improve control**. Check levels **after a seizure or if noncompliance is suspected**.
- **Targeted midpregnancy sonography** for anomalies in women on anticonvulsants.
- **Fetal well-being testing is not indicated** for uncomplicated epilepsy.
- **Breastfeeding:** limited data, but no obvious harm (e.g., no long-term cognitive effects).
- **Contraception:** COC failure ↑ with some anticonvulsants; **COCs substantially lower lamotrigine efficacy** → use more reliable methods.

4. Cerebrovascular Diseases

- Ischemic stroke = ↓ flow lasting longer than several seconds → infarction within minutes. Hemorrhagic stroke = bleeding into/around the brain (mass effect, blood toxicity, ↑ ICP).
- Nonpregnant women: mostly ischemic. Pregnant women: ~half ischemic, half hemorrhagic.**
- Incidence **10–40 per 100,000 births**, rising; contributes disproportionately to maternal mortality.
- US pregnancy-related deaths 2007–2016: **7.2% cerebrovascular accidents**, 7.8% hypertension. **9.8%** of deaths after 42 days postpartum were from stroke.
- Most events are associated with **hypertensive disorders or heart disease**.

Risk factors

- Timing:** most occur **intrapartum or postpartum**.

Series	Antepartum	Intrapartum	Postpartum
2850 pregnancy-related strokes (James, 2005)	~10%	~40%	~50%
145 ischemic strokes (Leffert, 2016)	45%	3%	53%

- Not pregnancy-related:** age, migraine, hypertension, obesity, diabetes, cardiac disease (endocarditis, prosthetic valves, **PFO**), smoking.
- Pregnancy-related:** hypertensive disorders, GDM, hemorrhage, cesarean.
 - Hypertensive disorders are by far the most common: one third** of strokes are associated with gestational hypertension; risk **3–8×**.
 - Stroke is the major cause of maternal death in preeclampsia.** General anesthesia may carry more risk than neuraxial.
 - Cesarean → 1.5× stroke risk** vs vaginal delivery. **Sepsis** ↑ postpartum stroke.
- Cerebral autoregulation is enhanced. **Cerebral blood flow ↓ 20%** from midpregnancy to term, but ↑ significantly with gestational hypertension (hyperperfusion is dangerous with vascular anomalies).

Table 63-4. Some Associated Disorders or Causes of Ischemic and Hemorrhagic Strokes During Pregnancy or the Puerperium

Ischemic Stroke	Hemorrhagic Stroke
Preeclampsia syndrome	Chronic hypertension
Arterial thrombosis	Preeclampsia syndrome
Venous thrombosis	Arteriovenous malformation
Lupus anticoagulant	Saccular aneurysm
Antiphospholipid antibodies	Angioma
Thrombophilias	Cocaine, methamphetamines
Migraine	Vasculopathy
Paradoxical embolus	
Cardioembolic	
Sickle hemoglobinopathy	
Arterial dissection	
Vasculitis	
Moyamoya disease	
Cocaine, amphetamines	

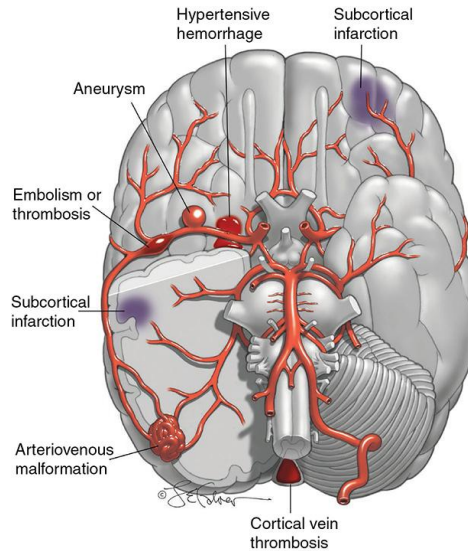


Figure 63-2. Stroke types in pregnancy: subcortical infarction (preeclampsia), hypertensive hemorrhage, aneurysm, MCA embolism/thrombosis, AVM, cortical vein thrombosis.

Ischemic stroke

- Acute occlusion or embolism of an intracranial vessel.
- **TIA**: reversible ischemia, symptoms **<24 h**; **~10% have a stroke within 1 year**.
- Presentation: **sudden severe headache, hemiplegia** or other deficits; occasionally seizures.
- **Workup**:
 - **Immediate assessment and CT**
 - Echocardiography; more cranial imaging (CT, MR, angiography)
 - Lipids (distorted by pregnancy)
 - **Antiphospholipid antibodies, lupus anticoagulant, other thrombophilias**, sickle-cell syndromes if indicated → these underlie **up to one third** of ischemic strokes in healthy young women
- Embolic stroke outcomes are favorable, similar to nonpregnant women. Thrombolysis and **mechanical thrombectomy** can be considered (experience sparse).

Type	Key points
Preeclampsia syndrome	A significant share of pregnancy ischemic strokes. Subcortical edema and petechial hemorrhage → infarction; usually shows as an eclamptic convulsion , occasionally a larger cortical infarct. Mimics: thrombotic microangiopathies and reversible cerebral vasoconstriction syndrome (postpartum angiopathy) → edema, necrosis, infarction, hemorrhage.
Cerebral embolism	Usually the middle cerebral artery . Diagnose only after excluding thrombosis and hemorrhage. Paradoxical embolism is uncommon although >1/4 of adults have a PFO (closure done in pregnancy, may not help). Cardioembolic causes: atrial fibrillation , valve lesions, MVP, mural thrombus, endocarditis, peripartum cardiomyopathy . Management: supportive + antiplatelet ; thrombolysis/anticoagulation controversial.
Cerebral artery thrombosis	Older patients, atherosclerosis (internal carotid); often preceded by TIAs. rtPA (alteplase) IV within 3 h if deficits and hemorrhage excluded on imaging. Main risk: hemorrhagic transformation. Can be used in pregnancy (few reports).
Cerebral venous thrombosis	6% of CVTs are pregnancy-associated; only 2% of pregnancy strokes . Greatest risk late pregnancy and puerperium . Lateral or superior sagittal sinus , usually puerperal , with preeclampsia, sepsis or thrombophilia/APA. Headache most common; deficits common; seizures in up to 1/3 . Can be misdiagnosed as postdural puncture headache . Dx: MR venography . Rx: anticonvulsants + heparin ; antimicrobials if septic; fibrinolysis only if anticoagulation fails. Mortality <10% (better than nonpregnant).

Recurrence

- Prior ischemic stroke → **low recurrence risk** unless a persistent cause exists; **not a contraindication to pregnancy** (441 women: 13 recurrences, only 2 pregnancy-associated).

- Prior CVT: 1 recurrence in 217 pregnancies; 5 noncerebral VTEs in 186. On prophylactic anticoagulation, no recurrence or bleeding, but **24%** had late obstetrical complications.
- **RCVS recurs in 9%** of subsequent pregnancies.
- No firm prophylaxis guidelines. Control hypertension and diabetes. **APS or certain cardiac conditions → prophylactic anticoagulation.** For most, **low-dose aspirin** throughout pregnancy is reasonable.

Hemorrhagic stroke

- Only **CT or MR** can separate it from ischemic stroke.
- Two categories: **intracerebral** and **subarachnoid** hemorrhage.

Intracerebral hemorrhage

- Bleeding into the parenchyma, usually from rupture of **Charcot-Bouchard aneurysms** (small vessels damaged by **chronic hypertension**).
- In pregnancy: **chronic hypertension with superimposed preeclampsia.**
- **Higher morbidity and mortality than SAH.** Sites: putamen, thalamus, white matter, pons, cerebellum.
- 28 women: **half died**, most survivors permanently disabled → **control gestational hypertension, especially systolic.**

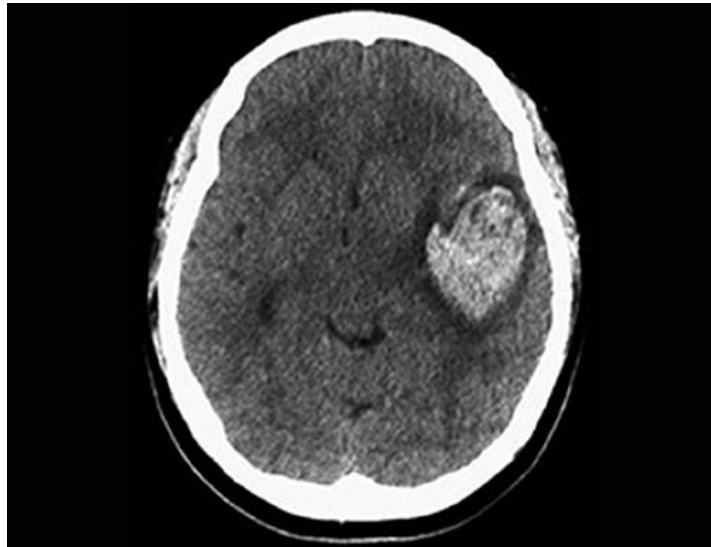


Figure 63-3. Noncontrast CT: large intraparenchymal hemorrhage in a woman with preeclampsia.

Subarachnoid hemorrhage

- Incidence **8.5 per 100,000 pregnancies.**
- Usually from an underlying **cerebrovascular malformation** in an otherwise normal woman.
- **Ruptured saccular ("berry") aneurysms cause 80% of all SAH.** Others: AVM, coagulopathy, angiopathy, venous thrombosis, infection, drugs, tumor, trauma.
- **Mortality lower in pregnancy: 8% vs 17%.**

Intracranial aneurysm

- Present in **1–2% of adults.** Rupture rate **~0.1% if <10 mm, ~1% if >10 mm.**
- Mostly **circle of Willis**; **multiple in 20%.**
- **Pregnancy does not ↑ rupture risk**, but high prevalence makes it the commonest SAH cause.
- Systematic review (50 aneurysms): **72% ruptured in pregnancy, 78% of these in the 3rd trimester.**
- **Symptoms: sudden severe headache** + visual change, cranial nerve palsy, focal deficit, altered consciousness. Also meningism, N/V, tachycardia, transient hypertension, low-grade fever, leukocytosis, proteinuria.
- **First test: noncontrast CT** (AHA), although MR may be superior.
- **Treatment:** bed rest, analgesia, neuro monitoring, **strict BP control.**

- Conservative treatment → rebleed **20–30% in the first month, then 3%/yr. Highest in the first 24 h**; rebleeding kills **70%**.
- Early repair after the sentinel bleed: craniotomy or **endovascular coiling** (coiling had **fewer complications than clipping** in pregnancy). Unruptured aneurysms: surgery → one third fewer complications than no treatment.
- **Remote from term**: repair without hypotensive anesthesia. **Near term: cesarean, then aneurysm repair.**
- Repaired aneurysm → vaginal delivery allowed if labor is "remote" from repair (some say **2 months**).
- **SAH survivor with no repair → no bearing down; favor cesarean.**

Arteriovenous malformations

- Congenital tangles of dilated arteries and veins **without capillaries** → AV shunting. Prevalence **0.01%**.
- Bleeds: **half subarachnoid, half intraparenchymal** with subarachnoid extension.
- In pregnancy: **83% ruptured during pregnancy or postpartum, >80% of these in the 2nd–3rd trimesters.** Bleeding may rise with GA but **not clearly more likely in pregnancy**; poor maternal outcome.
- **17%** of hemorrhagic strokes in one study. Parkland: 62 strokes in ~500,000 births over 35 years; **5 from AVMs.**
- **Associated aneurysm in up to 60%.**
- Rebleed if unrepaired: **6–20% in year 1, then 2–4%/yr. Mortality 10–20%.**
- Surgery decision is neurosurgical. **Unresected or inoperable AVM → favor cesarean.**

Memory hook: "Berry = Bulk of SAH (80%); Charcot-Bouchard = Chronic BP → intracerebral bleed."

5. Demyelinating or Degenerative Diseases

- **Demyelinating:** immune-mediated focal/patchy myelin destruction + inflammation. **Degenerative:** multifactorial, progressive neuronal death.

Multiple sclerosis

- Autoimmune demyelination; **second only to trauma** as a cause of neurological disability in mid-adulthood.
- **T-cell-mediated destruction of oligodendrocytes.** Familial with an environmental trigger. **Women 2:1**; onset in the **20s–30s. 0.03%** of California deliveries.

Type	Features
Relapsing (RMS)	Initial presentation in 90% . Unpredictable relapses, usually with full recovery; deficits accumulate over time.
Secondary progressive (SPMS)	Relapsing disease that then progresses after each relapse; probably all patients eventually.
Primary progressive (PPMS)	10%. Progressive disability from diagnosis.

- Classic findings: sensory loss, **optic neuritis**, weakness, paresthesias.
- Diagnosis: **MR imaging (multifocal white-matter plaques in >95%)** + CSF analysis. Plaque appearance does not predict treatment response.
- **Neuromyelitis optica spectrum disorder (NMOSD):** now a distinct disease; **antibodies to astrocytes** → spinal cord and optic nerve inflammation. **Pregnancy may worsen it.**

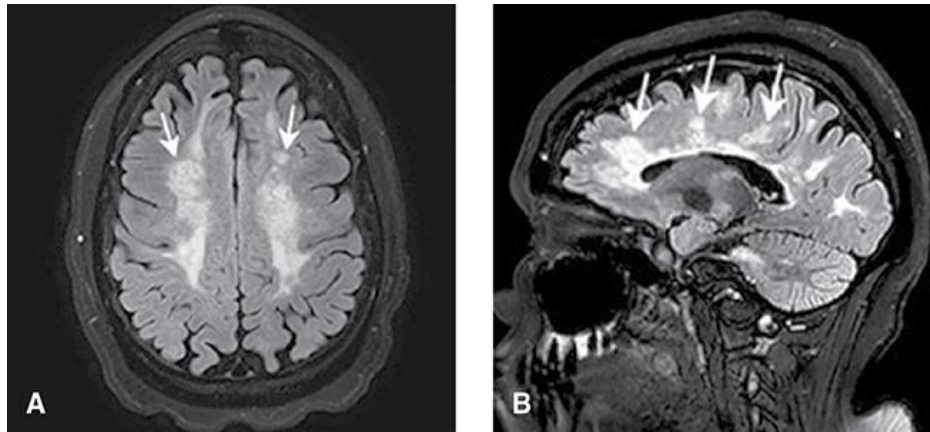


Figure 63-4. MS on MR imaging: (A) T2 bright white-matter lesions; (B) sagittal FLAIR periventricular lesions ("Dawson fingers").

MS in pregnancy

- **Fewer relapses in pregnancy, more postpartum** (↑ T-helper cells, ↑ T2/T1 ratio).

Period	Relapse rate (per year)
Before pregnancy	0.4
During pregnancy	0.26
After delivery	0.7

- Postpartum relapse predictors: high relapse rate before pregnancy, relapses during pregnancy, **high disability score**.
- **Breastfeeding has no effect** on postpartum relapse.
- Outcomes usually not affected. But: fatigue, **UTI with bladder dysfunction, autonomic dysreflexia if lesions at or above T6**.
- ↑ Induction, **longer second stage**, ↑ cesarean (inductions and elective surgery). Lower mean birthweight, but perinatal mortality similar. ↑ Infections and preterm delivery.

Management

- **Acute attack: IV methylprednisolone 500–1000 mg/d for 3–5 days** → oral prednisone for 2 weeks. **Plasma exchange** for fulminant steroid-resistant attacks.
- Symptomatic:

Symptom	Drug
Neurogenic pain	Carbamazepine, phenytoin or amitriptyline (after 1st trimester)
Spasticity	Baclofen
Bladder neck relaxation	α1-adrenergic blockers
Bladder contraction	Cholinergic (stimulate) / anticholinergic (inhibit)

- **Disease-modifying therapy:** interferons, glatiramer, fingolimod, dimethyl fumarate, natalizumab, ocrelizumab. Limited pregnancy data.
 - **Glatiramer:** not teratogenic (246 women).
 - **Natalizumab** in the 1st trimester → ↑ miscarriage.
 - **Fingolimod:** 6 malformations + 9 losses in 89 exposures and animal teratogenicity → **not recommended; contraception for 2 months after stopping**.
- **Postpartum relapse prevention: IVIG 0.4 g/kg/d × 5 days in weeks 1, 6 and 12.**

Myasthenia gravis

- Autoimmune neuromuscular disorder; **more common in women**, peaks in the **20s–30s**.
- **Anti-acetylcholine receptor antibodies in 80–90%**; seronegative patients often have **anti-MuSK** antibodies.

- **Weakness and fatigability** of facial, oropharyngeal, extraocular and limb muscles. **Deep-tendon reflexes preserved.**
- Early cranial involvement: **diplopia, ptosis**; trouble smiling, chewing, speaking. **Generalizes in 85%.**
- Exclude **hypothyroidism** (other autoimmune diseases coexist).

Crisis	Cause	Features
Myasthenic	Disease exacerbation	Severe weakness, can't swallow, respiratory muscle paralysis
Cholinergic	Excess anticholinesterase	Nausea, vomiting, weakness, abdominal pain, diarrhea

Management

- Manageable, not curable. **Pyridostigmine (Mestinon) is first line.**
- **Thymectomy** recommended but **postponed until after pregnancy** (done successfully in pregnancy for refractory cases).
- Refractory → glucocorticoids, azathioprine, cyclosporine.
- Rapid short-term improvement (surgery, crisis) → **high-dose IVIG or plasma exchange.**

Myasthenia and pregnancy

- **Highest risk in the first year after diagnosis** → postpone pregnancy until improvement is sustained.
- Close observation, rest, **treat infection promptly**. Continue corticosteroids/azathioprine if in remission.
- Course unpredictable; frequent admissions. **Up to one third worsen**, equally in all trimesters. Stable disease usually stays stable but **worsens in the first months postpartum.**
- Avoid **hypotension/hypovolemia** (can trigger crisis). The enlarging uterus can compromise breathing.
- **No significant adverse effect on pregnancy outcome.**
- ⚠️ **Preeclampsia: magnesium sulfate may precipitate a severe myasthenic crisis.** Phenytoin is also problematic but less so.
- **Labor:** smooth muscle unaffected → **normal labor**; oxytocin as usual; **cesarean for obstetrical indications only.**
 - Narcotics → respiratory depression; close observation.
 - ⚠️ **Avoid curariform drugs: magnesium sulfate**, general-anesthesia muscle relaxants, **aminoglycosides.**
 - **Neuraxial analgesia with amide-type local anesthetics**; regional preferred unless there is significant bulbar or respiratory involvement.
 - Weak pushing may need **operative vaginal delivery.**

Neonatal effects

- Maternal IgG anti-AChR and anti-MuSK antibodies **cross the placenta.**
- Poor fetal swallowing → **hydramnios.**
- **10–20% of neonates** have transient myasthenia: feeble cry, poor suckling, respiratory distress. Responds to anticholinesterases; **resolves in a few weeks.**

6. Neuropathies

- **Mononeuropathies** (single nerve; trauma, compression, entrapment) are relatively common in pregnancy. Childbirth can injure the **pudendal, obturator, femoral and common fibular** nerves (Ch 36).
- **Polyneuropathies:** axonal or demyelinating; linked to diabetes, toxins, drugs, genetic disease, rarely late syphilis.

Guillain-Barré syndrome

- Acute demyelinating polyradiculoneuropathy; **75% have evidence of recent infection: Campylobacter jejuni, CMV, Zika, EBV**; also surgery, immunizations.
- **Areflexic ascending paralysis** ± sensory loss; **autonomic dysfunction common.** Develops over **1–3 weeks.**

- Chronic inflammatory demyelinating polyneuropathy may be relatively common in young women.
- **Not more common in pregnancy**; same course.
- Management: **supportive** (VTE prophylaxis, pressure-ulcer prevention, enteral feeding). **One fourth need ventilation.**
- **IVIG or plasmapheresis** help if started **within 1–2 weeks** of motor symptoms, but **don't ↓ mortality**. Up to 10% relapse after improving → re-treat with **IVIG 2 g/kg over 5 days**.
- Most recover in months to a year. **Mortality 5%** (pulmonary complications, arrhythmias).

Bell palsy

- Acute facial nerve (mononeuropathic) paralysis; **female predominance**, possibly ↑ in pregnancy.
- Facial nerve inflammation, often **HSV or herpes zoster reactivation**; also Lyme disease/borreliosis.
- **Abrupt, painful onset; maximum weakness by 48 h**; ± hyperacusis and loss of taste.
- **Management**: facial massage, **eye protection**; **prednisone 1 mg/kg/d orally × 5 days** improves outcome. Adding valacyclovir/acyclovir is controversial.
- **Prognosis may be worse in pregnancy**: only **half** of pregnant women recovered satisfactorily at 1 year (vs **90%** nonpregnant).
- Poor prognostic markers: **bilateral**, recurrence in a later pregnancy, greater % nerve loss, faster loss.
- **5× ↑ gestational hypertension/preeclampsia**; no other adverse outcomes.



Figure 63-5. Bell palsy on the day of cesarean for twins; treated with prednisone + antiviral, almost resolved by 3 weeks.

Carpal tunnel syndrome

- **Median nerve compression; most frequent mononeuropathy in pregnancy.**
- Burning, numbness, tingling of the inner half of the hand(s); wrist pain, may extend to forearm and shoulder.
- **Bilateral in 80%; 10% severe denervation.** Incidence **7–43%**.
- Differential: **C6–C7 radiculopathy, de Quervain tendonitis**. Nerve conduction studies help.
- **Treatment: splint on a slightly flexed wrist at night.** Occasionally surgery or steroid injection.
- Symptoms persist in **>50% at 1 year, 1/3 at 3 years**.
- **Meralgia paresthetica: lateral femoral cutaneous nerve** compression → pain/burning/numbness of the **outer thigh**. Rx: corticosteroid injection where the nerve crosses the groin crease; oral **gabapentin**.

7. Spinal Cord Injury

- ~18,000 new US injuries/yr; mean age 42; 80% male. Women often have altered sexual function and transient hypogonadism, but **2%** reported pregnancy in the past year.

- **Two life-threatening problems:**

1. **Lesion above T10** → impaired cough, respiratory compromise, **aspiration pneumonia**. Consider PFTs; ventilation may be needed late in pregnancy/labor.
 2. **Lesion above T5–T6** → **autonomic dysreflexia**: stimuli below the lesion (reproductive/urinary tract or bowel distention/manipulation) → massive sympathetic discharge → **severe hypertension**, throbbing headache, flushing, sweating, **bradycardia or tachycardia**, arrhythmias, respiratory distress. **12 of 15** at-risk women had ≥1 episode in pregnancy; one case caused cerebral hemorrhage.
- Other complications: **pyelonephritis 30%**, lower UTI 32%, **pressure sores 9%**; constipation in >half; anemia.
 - **Labor**: contractions are unaffected → often easy, painless, even precipitous.
 - **Lesion below T12** → contractions felt normally.
 - **Lesion above T12** → risk of out-of-hospital delivery; **teach uterine palpation. Up to 20% deliver preterm.**
 - Options: tocodynamometry + weekly cervical exams from **28–30 wk**, or **elective admission at 36–37 wk** (Parkland).
 - **Vaginal delivery usually successful**; operative vaginal delivery often needed.
 - **Spinal/epidural analgesia to T10 at the start of labor prevents autonomic dysreflexia.**
 - Severe dysreflexia before epidural → remove the stimulus + **IV hydralazine or labetalol**.

8. Idiopathic Intracranial Hypertension (Pseudotumor Cerebri)

- ↑ **ICP without hydrocephalus**; CSF overproduction and/or underabsorption.
- Young, **obese** women or recent weight gain.
- Symptoms: **headache ≥90%**, **visual disturbance 70%**, papilledema (sight-threatening), stiff neck, back pain, **pulsatile tinnitus**, cranial nerve palsies.
- **Diagnostic criteria**:
 - **Opening pressure >25 cm H₂O**
 - **Normal CSF composition**
 - **Normal CT/MR imaging**
 - **Papilledema** (other criteria needed if absent)
 - No systemic disease
- Self-limited but **long-term vision loss is common**.
- **Treatment**: lower CSF pressure with **acetazolamide**, furosemide or topiramate. **Corticosteroids now rarely used**. Surgery: lumboperitoneal shunt, **optic nerve sheath fenestration**, venous sinus stenting.
- **In pregnancy**: unclear if pregnancy is a risk factor. Symptoms appear by midpregnancy and usually resolve postpartum.
 - **Pregnancy does not alter management**. Serial visual fields (16 women: field loss in 4, permanent in 1).
 - **Acetazolamide** for papilledema (12 treated successfully). **Repeated lumbar punctures** give temporary relief in pregnancy.
 - Complications stem from **obesity**, not IIH. **Delivery route by obstetrical indication; conduction analgesia is safe.**

9. Maternal Ventricular Shunts

- Ventriculoperitoneal, ventriculoatrial or ventriculopleural; outcomes usually satisfactory.
- **Partial obstruction common, especially late in pregnancy**. 17 pregnancies: neurological complications in 13. Headache 60%, N/V 35%, lethargy 30%, ataxia/gaze paresis 20% each.
- Most respond to conservative care. **CT shows acute hydrocephalus** → **tap or pump the shunt several times daily**; surgical revision if needed (may be emergent).

- **Endoscopic third ventriculostomy**: hole in the third-ventricle floor; successful in 5 gravidas.
- **Vaginal delivery preferred**. Conduction analgesia allowed **unless meningocele**.
- **Antibiotic prophylaxis if the peritoneum is entered** (cesarean or puerperal sterilization).

10. Maternal Brain Death

- Rare; few hospital policies address pregnancy.
- Life support and parenteral nutrition for **up to 20 weeks** awaiting delivery have been described ($\pm$ tocolysis and antimicrobials).
- 17 women in persistent vegetative state: 5 died after delivery; most others stayed vegetative.
- Profound ethical, financial and legal issues. **Perimortem cesarean** in some.

Quick-Recall: Disorder → Pregnancy Effect → Key Management

Disorder	Effect of pregnancy	Key management / pitfall
Migraine	Most improve ; new cases usually with aura	Acetaminophen/NSAID + antiemetic; triptans; no ergots ; no chronic NSAIDs late
Cluster headache	Unchanged	100% O ₂ , sumatriptan 6 mg SC; verapamil
Epilepsy	Seizures $\uparrow$ in 20–30%	Folic acid 4 mg ; monotherapy; avoid valproate ; midpregnancy anatomy scan
Ischemic stroke	Risk $\uparrow$, mostly peri/postpartum	CT; thrombophilia/APA workup; antiplatelet; rtPA ≤ 3 h if no bleed
Cerebral venous thrombosis	Puerperium	MR venography ; heparin
Aneurysmal SAH	Rupture risk not $\uparrow$; mostly 3rd trimester	Noncontrast CT; early coiling; near term $\rightarrow$ cesarean then repair
AVM	Not clearly $\uparrow$	Unresected $\rightarrow$ cesarean
Multiple sclerosis	Relapses $\downarrow$ in pregnancy, $\uparrow$ postpartum	Methylprednisolone; avoid fingolimod ; IVIG postpartum
Myasthenia gravis	$\frac{1}{3}$ worsen; worsen postpartum	Pyridostigmine; no MgSO₄ , no aminoglycosides; neonatal MG 10–20%
Guillain–Barré	Not $\uparrow$; same course	Supportive; IVIG/plasmapheresis within 1–2 wk
Bell palsy	Possibly $\uparrow$; worse recovery	Prednisone 1 mg/kg $\times$ 5 d; eye protection
Carpal tunnel	Common (7–43%)	Night wrist splint
Spinal cord injury	Painless, preterm labor	Epidural to T10 vs autonomic dysreflexia
IIH	Onset by midpregnancy	Acetazolamide; visual fields; LP
VP shunt	Obstruction late	Vaginal delivery; antibiotics if peritoneum entered

Key Numbers to Memorize

- Neurovascular disorders **>7%** of US maternal deaths; epilepsy **5%** of UK maternal deaths
- Pregnancy headache: $\frac{2}{3}$ **primary** (migraine >90%); $\frac{1}{3}$ **secondary** (>half hypertensive)
- Migraine: **15% women**; chronic = **≥ 15 d/mo for >3 mo**; stroke risk **16 $\times$** in pregnancy
- Prophylaxis: amitriptyline **10–175 mg**, propranolol **40–120 mg**, metoprolol **25–100 mg**; cluster: sumatriptan **6 mg SC**
- Folic acid **0.4 $\rightarrow$ 4 mg/d**; stop AEDs if seizure-free **≥ 2 yr**; seizure-free **9 mo $\rightarrow$ stable**; seizures $\uparrow$ **20–30%**

- Valproate **4–8×** (7–10% affected); phenytoin/phenobarbital **2–3×**; child's epilepsy risk **10%**; maternal death **10×**
- Stroke **10–40/100,000**; $\frac{1}{2}$ **ischemic** / $\frac{1}{2}$ **hemorrhagic**; cesarean **1.5×**; hypertensive women **3–8×**
- TIA → stroke **10% in 1 yr**; rtPA within **3 h**; CVT mortality **<10%**; RCVS recurrence **9%**
- SAH **8.5/100,000**; berry aneurysm **80%**; mortality **8% vs 17%**; aneurysm rupture **0.1% (<10 mm) vs 1% (>10 mm)**; rebleed kills **70%**
- AVM rebleed **6–20%** yr 1, then **2–4%/yr**; mortality **10–20%**; aneurysm in **60%**
- MS relapses **0.4 → 0.26 → 0.7/yr**; methylprednisolone **500–1000 mg × 3–5 d**; IVIG **0.4 g/kg × 5 d** at wk 1, 6, 12
- Myasthenia: anti-AChR **80–90%**; generalizes **85%**; neonatal MG **10–20%**
- GBS: infection **75%**; ventilation **25%**; mortality **5%**; re-treat IVIG **2 g/kg**
- Bell palsy: prednisone **1 mg/kg × 5 d**; recovery **50% vs 90%**; preeclampsia **5×**
- Carpal tunnel: bilateral **80%**; incidence **7–43%**
- Spinal cord: **T10** (respiratory; epidural level), **T5–T6** (dysreflexia), **T12** (labor sensation); preterm **20%**
- IIH: opening pressure **>25 cm H₂O**; headache **≥90%**; visual symptoms **70%**